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EVM

Gagan Singh(2025H1030051P)
Jenil Chandresh Gandhi(2025H1030060P)
Saunak Chakroborty(2025H1030P)

BITS Pilani
Pilani Campus

Why This Matters: Secure Offline Voting Under Real-World Threats

"Elections require absolute trust. Traditional systems face three critical vulnerabilities:"

Core challenge: correctness under tamper, reset, and power loss

1. **Vote tampering** – Physical access enables data modification
2. **Power failures** – Votes lost mid-recording undermine legitimacy
3. **Attack windows** – Delays in detecting breaches allow fraud completion

Goal: record votes exactly once in safe state, or transit to fail safe.

Project Context

Three Competing Constraints:

Constraint	Implication
Correctness	Every vote recorded exactly once, no duplicates, no loss
Security	Tamper detection ≤ 50 ms, lockdown ≤ 500 ms, irreversible
Reliability	Survive power loss, reset, sensor noise without corruption

Project Goal: **"Build a deterministic, fail-safe election controller that guarantees vote integrity under real-world faults—and prove it through rigorous verification."**

What the EVM Controller Owns (and What It Doesn't)

Inside system boundary

- Validate UART vote/control frames
- Enforce election state machine
- Record votes immutably in NVM
- Detect tamper and trigger fail-safe lockdown
- Generate audit/report data

Outside system boundary

- Voter authentication
- Cryptographic protocols
- Network/cloud communication
- GUI/policy decisions
- Physical security hardware design

Design assumptions

- Power/reset can happen anytime
- UART frames may be malformed/duplicated
- Sensor inputs can be noisy

Takeaway: **Clear system boundaries will enable focused design, stronger verification, and predictable fail-safe behavior.**

Requirements

We targeted these 4 key requirements; success means meeting these measurable criteria.

The system is considered **done** when all of the following are true:

Vote Integrity

- Every valid vote is recorded exactly once
- Votes persist across reset and power loss
- No duplication, loss, or modification of recorded votes

Tamper Safety

- All defined tamper events are detected
- System enters an irreversible safe state
- Previously recorded votes remain preserved

State Correctness

- Votes accepted only in Voting Active mode
- No undefined or unsafe state transitions
- System defaults to a safe non-voting state on faults

Determinism & Verification

- Behavior is repeatable across identical test runs
- Reset leads to consistent internal state
- Final election report matches stored vote data

When all measurable tests validating these conditions pass, the system is complete.

Success criteria

F-1: Valid vote recorded exactly once
F-3: Votes accepted only in Voting Active
S-1 / NF-5: Tamper detect ≤ 50 ms, lockdown ≤ 500 ms
S-2: No vote loss after sudden power failure
NF-1: Deterministic behavior for repeatable verification
NF-6: Complete audit trail for vote/state/tamper/admin events

If these requirements are met, the system is both technically correct and auditably trustworthy.

Secure EVM Controller – Architecture Overview

1. System Layer (Lifecycle & Recovery)

- **System** – Boot coordination, reset handling, subsystem initialization
- **Power Monitor** – Detects reset cause (power loss, WDT, software)
- **Logging Manager** – Immutable audit trail (tamper, state, reset)

Goal: Deterministic startup and fail-safe recovery.

2. Control Layer (State Authority)

- **Supervisor (Central State Machine)**
  - Owns election states:
    INIT → PRE_ELECT → VOTING → CLOSED
    TAMPER / ERROR (absorbing)
  - Validates all transitions
  - Blocks illegal operations

Goal: No voting outside authorized states.

3. Functional Layer (Voting & Security)

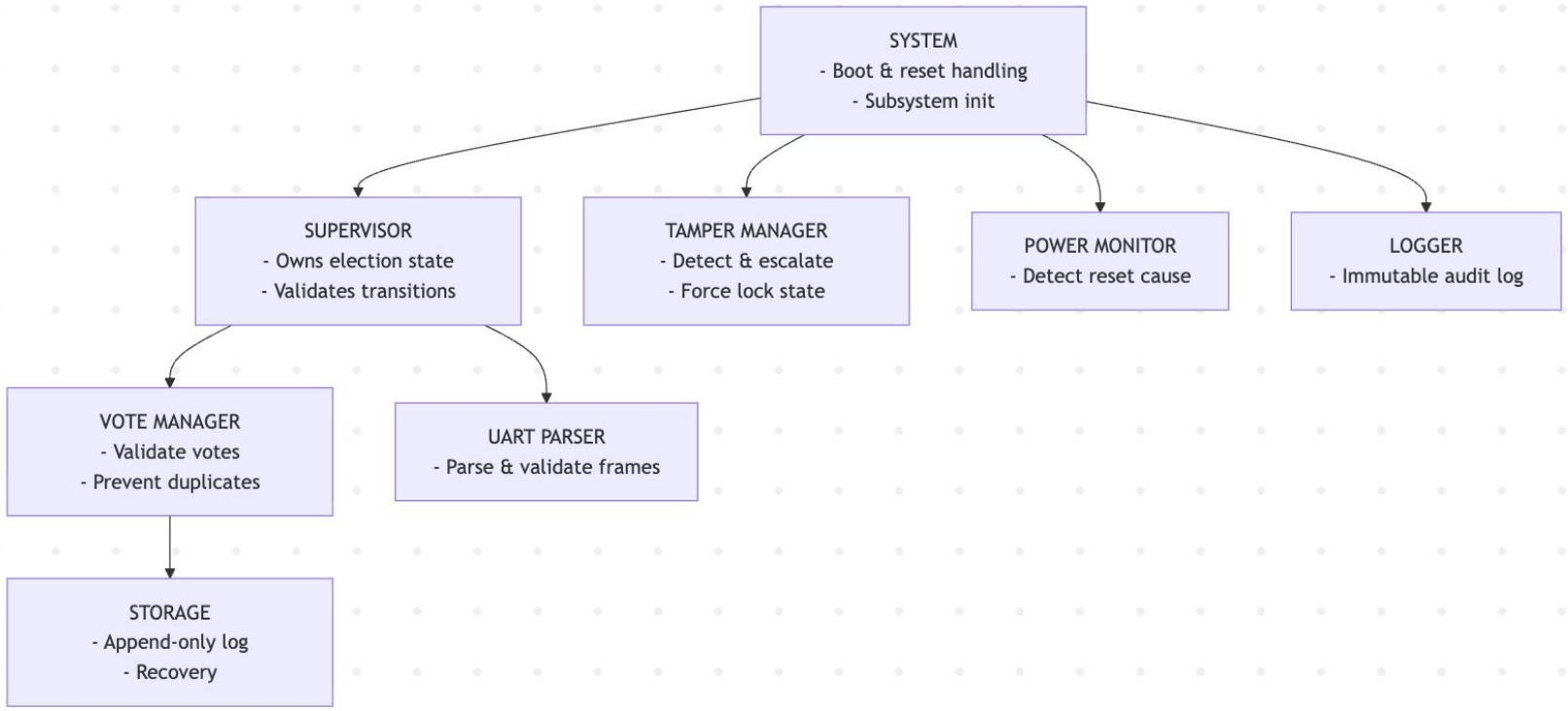
- **UART Parser** – Validates and extracts events
- **Vote Manager** – Validates votes, enforces exactly-once
- **Storage Manager** – Append-only persistent vote log
- **Tamper Manager** – Detects violations, forces lock state

Goal: Correct, immutable, tamper-resilient vote recording.

Architectural Guarantees

- Strict state control
- Append-only storage
- Irreversible tamper handling
- Safe power-loss recovery
- Full auditability

Secure EVM Controller – Architecture Overview



State behavior

Initialization → Pre-Election → Voting Active → Voting Closed → Audit

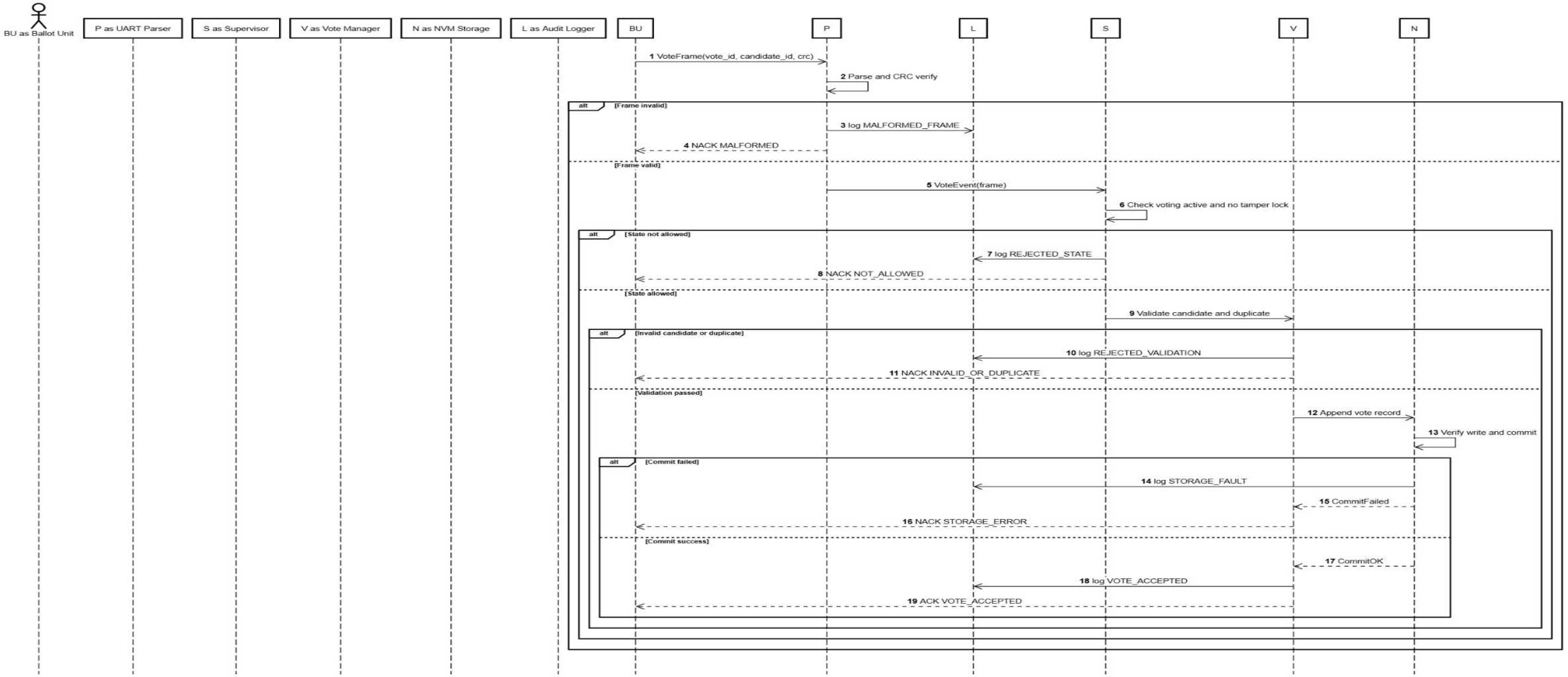
Tamper state reachable from operational states

Invalid transitions rejected and logged

Safe default on uncertainty: non-voting mode

state diagram

Deep dive into Vote Path: Exactly-Once Recording



Verification: How to Demonstrate System Correctness

Verification Strategy: 3 Test Layers

Layer	Purpose	Coverage
Unit Tests	Validate individual module logic in isolation	Parser, Vote Manager, Storage, Tamper Manager
Integration Tests	Verify module interactions and data flow	Supervisor ↔ Vote Manager ↔ Storage, Parser → Supervisor
System Tests	End-to-end scenarios matching real-world use	Full election cycle, power loss, tamper response

Multi-layer testing with requirement-to-test traceability ensures every critical behavior is verified and auditable.

Summary

What includes:

1. **Requirements** — 6 functional + security requirements (F-1 to F-3, S-1 to S-6, NF-1 to NF-6)
2. **Architecture** — 5 core modules: Supervisor, Vote Parser, Vote Manager, Storage, Tamper Manager, Audit Logger
3. **State Machine** — Election lifecycle: Initialization → Pre-Election → Voting Active → Voting Closed → Audit
4. **Use Cases** — Cast Vote, Detect Tamper Event
5. **Design Decisions** — Append-only NVM, interrupt-driven tamper response, layered validation gates
6. **Verification Plan** — Unit + integration + system test matrix with requirement traceability

Thank You